

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project — Python, MySQL, SQL & Power BI

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions. The workflow moves from raw data cleaning in Python, through structured business analysis in MySQL, to an interactive Power BI dashboard.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Customer demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- Missing Data: 37 values in the Review Rating column

3. Exploratory Data Analysis using Python

Data preparation and cleaning was performed in Jupyter Notebook using Pandas:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` and `df.describe()` to check structure and summary statistics.
- **Missing Data Handling:** Filled 37 missing Review Rating values using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake_case for readability.
- **Feature Engineering:** Created `age_group` (via quartile binning: Young Adult / Adult / Middle-aged / Senior) and `purchase_frequency_days` (mapped from purchase frequency labels).
- **Data Consistency Check:** Verified `discount_applied` and `promo_code_used` were identical; dropped the redundant `promo_code_used` column.
- **Database Integration:** Connected to MySQL using SQLAlchemy and loaded the cleaned DataFrame for SQL analysis.

Summary statistics (`df.describe()`):

Column	Count	Mean	Std	Min	50%	Max
Age	3900	44.07	15.21	18	44	70
Purchase Amount (USD)	3900	59.76	23.69	20	60	100
Review Rating	3863	3.75	0.72	2.5	3.8	5.0

Column	Count	Mean	Std	Min	50%	Max
Previous Purchases	3900	25.35	14.45	1	25	50

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in MySQL to answer key business questions:

1. Revenue by Gender — total revenue generated by male vs. female customers.

Gender	Revenue (USD)
Female	75,191
Male	157,890

2. High-Spending Discount Users — customers who used a discount but still spent above the average purchase amount (\$59.76).

Result: **839 customers** matched this condition (out of 3,900).

3. Top 5 Products by Rating — products with the highest average review ratings.

Item Purchased	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison — average purchase amount by shipping type.

Shipping Type	Avg Purchase (USD)
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers — average spend and total revenue.

Subscription Status	Customers	Avg Spend	Total Revenue
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

6. Discount-Dependent Products — highest percentage of discounted purchases.

Item Purchased	Discount %
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation — New (≤ 1 previous purchase), Returning (2–4), Loyal (5+).

Customer Segment	Number of Customers
Loyal	3,563
Returning	254
New	83

8. Top 3 Products per Category — most purchased products within each category.

Category	Item Purchased	Total Orders	Rank
Accessories	Jewelry	171	1
Accessories	Belt	161	2
Accessories	Sunglasses	161	3
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

9. Repeat Buyers & Subscriptions — subscription split among customers with more than 5 previous purchases.

Subscription Status	Repeat Buyers (>5 purchases)
No	2,518

Subscription Status	Repeat Buyers (>5 purchases)
Yes	958

10. Revenue by Age Group — total revenue contribution of each age group.

Age Group	Total Revenue (USD)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

5. Dashboard in Power BI

An interactive Power BI dashboard was built on the cleaned dataset to present these insights visually, with slicers for subscription status, gender, category, and shipping type.



6. Business Recommendations

- **Boost Subscriptions** — Only 27% of customers are subscribed; promote exclusive subscriber-only discounts or early access to drive sign-ups.
- **Customer Loyalty Programs** — With 3,563 customers already in the Loyal segment, introduce tiered rewards to retain them and convert Returning customers (254) into Loyal ones.
- **Review Discount Policy** — Hats, Sneakers, and Coats rely on discounts for ~48-50% of their sales; assess whether this is eroding margins versus driving incremental volume.
- **Product Positioning** — Highlight top-rated products (Gloves, Sandals, Boots, Hat, Skirt) in marketing campaigns to leverage strong customer satisfaction.
- **Targeted Marketing** — Young Adults generate the highest revenue (\$62,143); tailor campaigns and express-shipping promotions toward this segment.